Basilicata-Akhtar Syndrome (MRXSBA): A Comprehensive Disease Characterization

Target disease: Basilicata-Akhtar syndrome (MONDO:0026730; OMIM #301032) **Causal gene: MSL3** (MSL complex subunit 3; HGNC:7370; Xp22.2) **Category:** Mendelian, X-linked dominant neurodevelopmental chromatinopathy

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Summary

Basilicata-Akhtar syndrome — also called **MRXSBA** (intellectual developmental disorder, X-linked syndromic, Basilicata-Akhtar type) and **MSL3-related disorder** — is an ultra-rare X-linked dominant neurodevelopmental disorder caused by predominantly **de novo loss-of-function variants in MSL3**, the gene encoding a chromatin-reader subunit of the **MSL (male-specific lethal) histone acetyltransferase complex**. The MSL complex, whose catalytic subunit is MOF/KAT8, deposits **histone H4 lysine 16 acetylation (H4K16ac)**, a mark that decompacts chromatin and promotes transcription of active gene bodies. Pathogenic MSL3 variants impair assembly of this complex and reduce genome-wide H4K16ac in patient cells, dysregulating developmental gene programs that drive neurogenesis, morphogenesis, and cell migration. This places the disorder squarely among the "chromatinopathies" — Mendelian disorders of the epigenetic machinery.

Clinically, affected males and females present from infancy with **global developmental delay and intellectual disability (ranging from mild to profound)**, **muscular hypotonia** that may evolve into **progressive spasticity and gait disturbance**, **autism spectrum disorder (~50%)**, **feeding/gastrointestinal problems**, **hearing impairment**, **macrocephaly**, a **recognizable facial gestalt** (smooth philtrum, downturned mouth, deeply set eyes, low-set ears, tented thin upper lip, wide nasal ridge, short neck), and a highly consistent neuroimaging finding of **inferior cerebellar vermis hypoplasia** on brain MRI. As of 2025, approximately **42 cases** have been documented, affecting both sexes roughly equally. Diagnosis is molecular, via whole-exome or whole-genome sequencing; pathogenic variants cluster in the **terminal eight exons (exons 6–13) encoding the MRG domain**.

There is **no curative, disease-specific therapy**; management is supportive and multidisciplinary (physical/occupational/speech therapy, developmental and autism support, audiology, gastroenterology, and surveillance imaging). Mechanism-based and targeted leads include **HDAC inhibitors**, which restore acetylation balance and rescue molecular/cellular phenotypes in patient cells (experimental); **recombinant human growth hormone (rhGH)** for co-occurring growth hormone deficiency/short stature; and **carbidopa-levodopa** for parkinsonian motor features reported in an adult patient. Survival into at least the fourth decade is documented, and no disease-specific mortality or reduced life-expectancy data have been published.

Key Findings

Finding 1 — MRXSBA is an X-linked neurodevelopmental disorder caused by MSL3 loss-of-function variants

Basilicata-Akhtar syndrome is defined by loss-of-function variants in **MSL3**, an X-chromosomal gene at **Xp22.2**. The defining 25-individual cohort (15 males, 10 females) reported by Brunet et al. (2021) catalogued a broad spectrum of variant types — **10 nonsense, 6 frameshift, 4 splice-site, 3 missense, 1 in-frame deletion, and 1 multi-exon deletion** — the great majority proven to be *de novo*. Critically, these variants cluster in the **terminal eight exons**, a pattern the authors interpret mechanistically: truncating variants in the first five exons may be compensated by an alternative MSL3 transcript, so only variants affecting the later exons produce disease.

> "We identified multiple variant types in MSL3 (ten nonsense, six frameshift, four splice site, three missense, one in-frame-deletion, one multi-exon deletion), most proven to be de novo, and clustering in the terminal eight exons suggesting that truncating variants in the first five exons might be compensated by an alternative MSL3 transcript." — PMID: 33173220

The core clinical phenotype comprises **developmental delay and intellectual disability (mild to severe)**, with **autism spectrum disorder, muscle tone abnormalities, and macrocephaly** common, alongside **hearing impairment and gastrointestinal problems**, and a consistent MRI finding of **cerebellar vermis hypoplasia**.

> "The main clinical findings comprised developmental delay and intellectual disability ranging from mild to severe. Autism spectrum disorder, muscle tone abnormalities, and macrocephaly were common as well as hearing impairment and gastrointestinal problems. Hypoplasia of the cerebellar vermis emerged as a consistent magnetic resonance image (MRI) finding." — PMID: 33173220

A 2025 case report and review confirms the small but growing case count and the predominance of de novo loss-of-function variants:

> "To date, only 42 cases have been documented, affecting both males and females, with most variants being de novo and loss-of-function." — PMID: 40767387

Finding 2 — MSL3 encodes a reader subunit of the MSL histone acetyltransferase complex that targets global H4K16 acetylation

The MSL complex comprises the catalytic subunit **MOF/KAT8** plus **MSL1, MSL2, and MSL3**, and it catalyzes **histone H4 lysine 16 acetylation (H4K16ac)** — a modification that decompacts the chromatin fiber and marks transcriptionally active gene bodies. Recent cryo-EM and biochemical work shows the MSL subunits act cooperatively as a "molecular ruler" conferring residue specificity, with recognition of H3K36 trimethylation accelerating catalysis and thereby coupling MSL activity to active transcription.

> "The cooperative interplay of MSL subunits acts as a molecular ruler conferring specificity, while recognition of H3K36 trimethylation accelerates enzymatic activity, linking MSL function to transcriptionally active gene bodies." — PMID: 42880250

MSL3 contributes a **chromo/MRG domain** to the complex. Recent work establishes that the **MRG domain is intolerant to loss-of-function variants and is responsible for MSL3 function**:

> "Our data support the notion that the MRG domain of MSL3 is intolerant to loss-of-function variants and responsible for MSL3 function." — PMID: 41216464

The downstream consequences of losing MSL-complex activity in the nervous system are demonstrated in a conditional-knockout mouse model of the partner subunit Msl2:

> "Msl2-deficient brains exhibit significantly reduced levels of histone H4 lysine 16 acetylation (H4K16ac), particularly at promoter regions. Integrated RNA-seq, ChIP-seq, and ATAC-seq analyses revealed widespread transcriptional downregulation and loss of chromatin accessibility at NDD-related genes." — PMID: 42168661

Finding 3 — MSL3 variants reduce H4K16ac in vivo, dysregulate morphogenesis/cell-migration programs, and HDAC inhibitors rescue patient-cell phenotypes

The syndrome-defining paper (Basilicata et al., 2018) established the molecular pathology directly in patient cells. Pathogenic MSL3 variants impair MSL complex assembly and activity, with a pronounced loss of H4K16ac in vivo. Patient-derived cells show global transcriptome alterations in pathways governing **morphogenesis and cell migration**, and treatment with **histone deacetylase inhibitors (HDACi)** rebalances acetylation and alleviates molecular and cellular phenotypes — providing both mechanistic proof and a candidate therapeutic strategy.

> "MSL3 mutations affect MSL complex assembly and activity, accompanied by a pronounced loss of H4K16ac levels in vivo. Patient-derived cells display global transcriptome alterations of pathways involved in morphogenesis and cell migration. Finally, we use histone deacetylase inhibitors to rebalance acetylation levels, alleviating some of the molecular and cellular phenotypes of patient cells." — PMID: 30224647
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> "Clinical features of the syndrome include global developmental delay, progressive gait disturbance, and recognizable facial dysmorphism." — PMID: 30224647

Finding 4 — MSL3 is evolutionarily conserved and reads H3K36me3 to direct MSL complex spreading

MSL3's chromodomain reads nucleosomes marked by **H3K36me3** and directs the **targeting/spreading** step of the MSL complex across active genes. Chromodomain mutants retain binding to chromatin entry sites but lose the spreading pattern and lose preferential affinity for H3K36me3-marked nucleosomes.

> "MSL3 chromodomain mutants retain binding to chromatin entry sites but show a clear disruption in the full pattern of MSL targeting in vivo, consistent with a loss of spreading. Furthermore, when compared to wild type, chromodomain mutants lack preferential affinity for nucleosomes containing H3K36me3 in vitro." — PMID: 19029895

The machinery is deeply conserved: the budding-yeast protein **Eaf3p** is a homolog of MSL3 and a stable component of the NuA4 H4 acetyltransferase complex, linking yeast, flies (*Drosophila* msl-3, a dosage-compensation protein), and mammals.

> "Eaf3p (Esa1p-associated factor-3 protein), a yeast protein homologous to the Drosophila dosage compensation protein MSL3, is also a stable component of the NuA4 complex." — PMID: 11036083

Finding 5 — Detailed phenotype, locus, and inheritance of MRXSBA

MSL3 maps to **Xp22.2**. The clinical picture (OMIM #301032; PMID: 30224647, PMID: 33173220) comprises global developmental delay from infancy, feeding difficulties, muscular hypotonia, poor or absent speech, intellectual disability (mild–severe), autism spectrum disorder, macrocephaly, hearing impairment, gastrointestinal problems, and cerebellar vermis hypoplasia on MRI. Most patients are ambulatory but with an **unsteady gait and/or progressive spasticity**. A distinctive **facial gestalt** (smooth philtrum, downturned corners of the mouth, deeply set eyes, low-set ears, tented/thin upper lip, wide nasal ridge, short neck) plus mild distal skeletal anomalies is recognizable. Inheritance is **X-linked dominant**, with males and females affected similarly; nearly all variants are *de novo*, with one reported sib pair attributed to suspected parental somatic/germline mosaicism. At the molecular level, mutant MSL3 proteins lose interaction with **MOF (KAT8) and MSL1**, with bulk reduction of H4K16ac in patient-derived cells.

Finding 6 — Quantitative phenotype frequencies and variant distribution

The Brunet et al. (2021) cohort (n=25; 15 M, 10 F) provides frequency data suitable for knowledge-base annotation:

Phenotype	Frequency	HPO suggestion
Developmental delay / intellectual disability	24/25 (essentially all; one isolated speech delay)	HP:0001263 / HP:0001249
Autism spectrum disorder	10/20 (50%)	HP:0000729
Muscular hypotonia	21/25	HP:0001252
Seizures	4/23 (~17%, well-controlled)	HP:0001250
Cerebellar vermis hypoplasia (MRI)	Consistent finding	HP:0001320
Macrocephaly	Common	HP:0000256
Hearing impairment	Common	HP:0000365
Gastrointestinal problems	Common	HP:0011024
Genitourinary problems	Reported	HP:0000119
Vision abnormalities	Reported	HP:0000504
Ataxia	6/25	HP:0001251
Dystonia	5/25	HP:0001332

Variant distribution: **10 nonsense, 6 frameshift, 4 splice-site, 3 missense, 1 in-frame deletion, 1 multi-exon deletion — ALL within the last 8 exons (exons 6–13) encoding the MRG domain.** Inheritance: 21 proven *de novo*, 3 presumed *de novo*, plus 1 case of low-level somatic mosaicism in a phenotypically normal mother.

> "most proven to be de novo, and clustering in the terminal eight exons suggesting that truncating variants in the first five exons might be compensated by an alternative MSL3 transcript" — PMID: 33173220

Finding 7 — Diagnosis, management, and treatment

Diagnosis is molecular, via whole-exome or whole-genome sequencing with Sanger confirmation; pathogenic MSL3 variants cluster in exons 6–13 (MRG domain). MSL3 has two functional domains: the **MRG domain** (responsible for MSL complex formation / protein interactions with MOF-KAT8 and MSL1) and the **chromo domain** (chromatin targeting via H3K36me3).

No disease-specific curative therapy exists. Management is supportive and multidisciplinary: physical/occupational therapy (hypotonia, gait, progressive spasticity), speech therapy (poor/absent speech), developmental/behavioral and autism support, audiology (hearing impairment), gastroenterology/feeding and constipation management, and brain MRI surveillance (cerebellar vermis hypoplasia). Brunet et al. (2021) proposed detailed surveillance guidelines organized by affected organ system.

Emerging / targeted options:

- **Recombinant human growth hormone (rhGH)** improved height over >10-year follow-up in individuals with MRXSBA and growth hormone deficiency:

> "Over 10 years of clinical follow-up indicates that rhGH treatment could be an option to improve the height in individuals with MRXSBA and GHD." — PMID: 41216464

- **Carbidopa-levodopa** improved parkinsonian motor features in an adult patient:

> "the individual is receiving treatment with carbidopa-levodopa with improvement in bradykinesia and camptocormia" — PMID: 40767387

- **HDAC inhibitors** (experimental, mechanism-directed):

> "we use histone deacetylase inhibitors to rebalance acetylation levels, alleviating some of the molecular and cellular phenotypes of patient cells" — PMID: 30224647

Finding 8 — Model organisms implicate MSL/H4K16ac in neural progenitor priming and differentiation

Human cellular models (patient lymphoblastoid/fibroblast lines; HEK293 transfection) demonstrate that mutant MSL3 loses MOF/MSL1 interaction and reduces bulk H4K16ac, with transcriptome dysregulation (predominantly downregulation) of developmental-pathway genes. The closest in vivo model is the **nervous-system-specific Msl2 conditional-knockout mouse**, which recapitulates key disease biology:

> "Using a nervous system-specific Msl2 conditional knockout mouse model, we demonstrate that Msl2 deficiency leads to impaired social novelty recognition, learning deficits, and spatial memory impairments. These behavioral abnormalities are characterized by disrupted neocortical lamination, compromised proliferation capacity and differentiative ability of neural progenitor cells, and reduced neuronal migration." — PMID: 42168661
>
> "Patient-derived cells display global transcriptome alterations of pathways involved in morphogenesis and cell migration." — PMID: 30224647

Supporting invertebrate/fungal models include *Drosophila* msl-3 (dosage compensation; chromodomain H3K36me3 reader, PMID: 19029895) and *S. cerevisiae* EAF3/NuA4 (PMID: 11036083).

Finding 9 — Identifiers, gene constraint, severity distribution, and prognosis

Gene identifiers: MSL3 — HGNC:7370; NCBI Gene 10943; Ensembl ENSG00000005302; UniProt Q8N5Y2; gene OMIM *300609; MANE Select NM_078629.4 / ENST00000312196.10; cytoband Xp22.2; GRCh38 chrX:11,758,159–11,775,772.

Disease identifiers: OMIM #301032 (MRXSBA); MONDO:0026730; DOID:0111838; MedGen C5231394.

Gene constraint (gnomAD v4): pLI = 1, LOEUF = 0.18 — MSL3 is **highly intolerant to loss-of-function variation**, consistent with haploinsufficiency as the disease mechanism.

Severity distribution (Brunet 2021): global developmental delay in 24/25 (isolated speech delay in 1); among 15 formally tested for intellectual disability — mild (5), moderate (3), severe (1), profound (1), borderline (1); hypotonia in 21; movement disorders including dystonia (5) and ataxia (6); seizures in 4/23 (well controlled).

> "The main clinical findings comprised developmental delay and intellectual disability ranging from mild to severe." — PMID: 33173220

Prognosis: Patient ages ranged from 16 months to 24 years in the defining cohort, plus a 30-year-old reported case, indicating survival into adulthood. No disease-specific mortality or reduced life-expectancy data have been published.

> "We present a case of a 30-year-old male with BAS, exhibiting developmental delay, intellectual disability, hypotonia, motor disturbances, and dysmorphic features." — PMID: 40767387

Example ClinVar pathogenic variants: NM_078629.4:c.1372C>T p.(Arg458Ter); c.1293G>A p.(Trp431*); c.1055C>T p.(Thr352Met).

Section-by-Section Report

1. Disease Information

Overview. Basilicata-Akhtar syndrome (MRXSBA; MSL3-related disorder) is an ultra-rare, X-linked dominant syndromic neurodevelopmental disorder caused by variants in **MSL3**. It is best understood as a **chromatinopathy** — a Mendelian disorder of the epigenetic machinery — because MSL3 is a subunit of the histone-acetyltransferase complex that deposits H4K16ac.

Key identifiers. OMIM #301032; MONDO:0026730; DOID:0111838; MedGen C5231394; gene OMIM *300609 (MSL3). An Orphanet code and ICD-11 mapping were not individually confirmed in this investigation; the disorder falls under broad categories for syndromic X-linked intellectual disability. MeSH does not carry a dedicated descriptor; it is indexed under intellectual disability / chromatin-related mechanisms.

Synonyms. Basilicata-Akhtar syndrome; MRXSBA; intellectual developmental disorder, X-linked syndromic, Basilicata-Akhtar type; MSL3-related disorder; X-linked dominant MSL3 disorder.

Information source. Derived from aggregated disease-level resources (OMIM, MONDO) and small published patient cohorts/case reports (individual patient data), not from large EHR datasets.

2. Etiology

Primary cause. Genetic — predominantly **de novo loss-of-function variants in MSL3** (nonsense, frameshift, splice-site; plus select missense and in-frame/multi-exon deletions), clustering in exons 6–13 (MRG domain).

Genetic risk factors. The causal variant itself is the risk factor; MSL3 is highly LoF-intolerant (pLI=1, LOEUF=0.18). No susceptibility loci or modifier genes are established. A proposed genotype–phenotype modifier is the **existence of an alternative MSL3 transcript** that may compensate for truncating variants in the first five exons, explaining why disease variants cluster in the terminal exons (PMID: 33173220).

Environmental risk / protective factors. None identified — consistent with a monogenic, de novo etiology. **Gene–environment interactions:** none established.

3. Phenotypes

See the frequency table under Finding 6. Phenotype types span clinical signs (hypotonia, spasticity, dysmorphism), physical manifestations (macrocephaly, cerebellar vermis hypoplasia), behavioral changes (ASD, intellectual disability), and functional impairments (poor/absent speech, feeding difficulty). Onset is **neonatal/infantile**, severity is **variable (mild to profound)**, and the course is best described as a **stable-to-slowly-progressive** developmental disorder with some **progressive motor features** (spasticity, gait disturbance, and parkinsonism in an adult). Quality-of-life impact is substantial given intellectual disability, communication impairment, and motor/feeding needs, though no formal EQ-5D/SF-36 data exist for this rare disease.

4. Genetic / Molecular Information

Causal gene: MSL3 (HGNC:7370; Xp22.2; gene OMIM *300609). Variant classes: nonsense (most common), frameshift, splice-site, missense, in-frame deletion, multi-exon deletion. Classification: pathogenic/likely pathogenic per ACMG (loss-of-function in a LoF-intolerant gene). Allele frequency: absent/ultra-rare in gnomAD (de novo). Origin: germline de novo (one documented maternal low-level somatic mosaicism). Functional consequence: loss of function / haploinsufficiency — mutant MSL3 loses MOF(KAT8)/MSL1 interaction, impairing MSL complex*

*assembly and reducing H4K16ac. Modifier genes: none confirmed (alternative transcript hypothesis noted). Epigenetic consequence: reduced H4K16ac genome-wide, especially at promoters. Chromosomal abnormalities:** one multi-exon deletion documented; otherwise point/small variants.

5. Environmental Information

Not applicable. No environmental, lifestyle, or infectious contributors are implicated; the disorder is monogenic and de novo.

6. Mechanism / Pathophysiology

Ordered causal chain:

1. A **de novo loss-of-function variant in MSL3** (clustering in exons 6–13, MRG domain) **leads to** a truncated/non-functional MSL3 protein (haploinsufficiency).
2. Loss of functional MSL3 **results in** failure of the MRG domain to mediate protein–protein interactions with **MOF (KAT8) and MSL1**, impairing **MSL complex assembly**.
3. Impaired complex assembly, together with loss of the MSL3 **chromodomain's H3K36me3-reading/spreading** function, **leads to** a pronounced **genome-wide reduction of H4K16ac**, particularly at promoters (demonstrated in patient cells and in the Msl2-cKO mouse).
4. Reduced H4K16ac **results in** chromatin that is less decompacted and less accessible, **leading to widespread transcriptional downregulation** of developmental genes, including programs for **morphogenesis and cell migration** and NDD-related genes.
5. (Branch — neurodevelopment) Transcriptional dysregulation **leads to** impaired **neural progenitor proliferation and differentiation, reduced neuronal migration, and disrupted neocortical lamination** (inferred from the Msl2-cKO mouse), **resulting in** developmental delay, intellectual disability, and autism-like behavior.
6. (Branch — cerebellum) Disrupted neurodevelopment **results in inferior cerebellar vermis hypoplasia** (consistent MRI finding; mechanistic link inferred).
7. (Branch — motor) Downstream circuit dysfunction **leads to** hypotonia that may evolve into **progressive spasticity, ataxia, dystonia, and (in adulthood) parkinsonism**.

Molecular pathways: chromatin modification / histone acetylation (not a classic signaling cascade). **Cellular processes (GO):** histone H4-K16 acetylation (GO:0043984), chromatin organization (GO:0006325), regulation of transcription (GO:0006355), neuron migration (GO:0001764), neural progenitor proliferation/differentiation. **Protein dysfunction:** loss of

function / disrupted complex assembly. **Cell types (CL):** neural progenitor cell (CL:0011020), neuron (CL:0000540), cerebellar neurons. **Epigenetic change:** reduced H4K16ac. **Subcellular (GO CC):** nucleus (GO:0005634), MSL complex (GO:0072487), chromosome.

7. Anatomical Structures Affected

Organ level: brain (primary) — **UBERON:0000955**; cerebellum/cerebellar vermis — **UBERON:0002037** / **UBERON:0004086**; cerebral cortex (neocortex) — **UBERON:0001950**. Secondary/body-system involvement: central nervous system, auditory system (hearing impairment, UBERON:0001690), gastrointestinal system (UBERON:0001555), genitourinary system, musculoskeletal (hypotonia/spasticity, mild distal skeletal anomalies), craniofacial (dysmorphism, macrocephaly). **Tissue/cell level:** neural tissue; neural progenitor cells and neurons. **Subcellular:** nucleus/chromatin. **Lateralization:** bilateral/symmetric (cerebellar vermis is a midline structure).

8. Temporal Development

Onset: congenital/infantile; insidious, non-acute. **Progression:** largely stable developmental disorder with slowly progressive motor features (spasticity, gait disturbance; parkinsonism reported in a 30-year-old). **Course:** chronic, lifelong. **Critical period:** early neurodevelopment (progenitor proliferation/migration/lamination) is the mechanistically vulnerable window. No remission occurs; seizures, when present, are typically well-controlled.

9. Inheritance and Population

Epidemiology: ultra-rare; ~42 published cases as of 2025; precise prevalence/incidence undefined. **Inheritance:** X-linked dominant; males and females affected similarly. **Penetrance:** appears high for de novo pathogenic variants. **Expressivity:** variable (ID mild to profound). **Mosaicism:** one maternal low-level somatic mosaicism case (phenotypically normal carrier mother). **De novo predominance:** 21/25 proven + 3 presumed de novo in the defining cohort. **Founder effects/consanguinity:** not applicable (de novo). **Sex ratio:** roughly equal — a notable feature for an X-linked disorder. **Age distribution:** 16 months to 30+ years reported.

10. Diagnostics

Genetic testing is the diagnostic cornerstone: **whole-exome or whole-genome sequencing**, with Sanger confirmation; single-gene MSL3 testing or NDD/ID gene panels including MSL3 are appropriate. Chromosomal microarray may detect the rare multi-exon deletion. **Imaging:** brain MRI is key — **inferior cerebellar vermis hypoplasia** is the consistent and

distinguishing finding. **Audiology** for hearing impairment. **No specific biochemical biomarker** exists; H4K16ac reduction is a research assay, not a clinical test. **Differential diagnosis:** other chromatinopathies / Mendelian disorders of the epigenetic machinery presenting with DD/ID, hypotonia, and facial dysmorphism (e.g., KAT6A/KAT6B, KANSL1/Koolen-de Vries, and other MSL/NSL-pathway disorders), distinguished by molecular testing. **Screening:** no newborn/population screening; **cascade testing** rarely informative given de novo origin, but relevant when parental mosaicism is suspected.

11. Outcome / Prognosis

Survival into at least the fourth decade is documented; **no disease-specific mortality or reduced life-expectancy data** have been published. Morbidity is driven by **intellectual disability, communication impairment, motor dysfunction (hypotonia→spasticity, ataxia, dystonia, parkinsonism), feeding/GI issues, and hearing impairment**. Recovery is not expected; management is lifelong supportive care. **Prognostic factors:** degree of intellectual disability and motor involvement. No validated prognostic biomarkers.

12. Treatment

No curative therapy. Supportive, multidisciplinary care is standard: physical/occupational therapy, speech therapy, developmental/behavioral and autism support, audiology, gastroenterology/feeding and constipation management, orthopedic/neurology input for spasticity and movement disorders, and MRI surveillance. **Targeted/emerging: rhGH** (NCIT: Recombinant Human Growth Hormone) for co-occurring GH deficiency/short stature over >10-year follow-up; **carbidopa-levodopa** (NCIT: Levodopa/Carbidopa) for parkinsonian features in an adult; **HDAC inhibitors** (NCIT: Histone Deacetylase Inhibitor) as an experimental, mechanism-directed strategy to restore H4K16ac balance (demonstrated in patient cells only — not a clinical therapy). Anti-seizure medication controls seizures well when present. **Pharmacogenomics / gene therapy / RNA therapy:** none established.

13. Prevention

Primary prevention is not applicable for de novo variants. **Genetic counseling** is central: recurrence risk is generally low (de novo), but **parental mosaicism** (as in the one documented maternal somatic-mosaicism case) raises recurrence risk and warrants discussion; **prenatal/preimplantation genetic testing** is possible when a familial variant or mosaicism is identified. **Tertiary prevention** (preventing complications) is achieved through surveillance of hearing, GI, growth, and neurologic status.

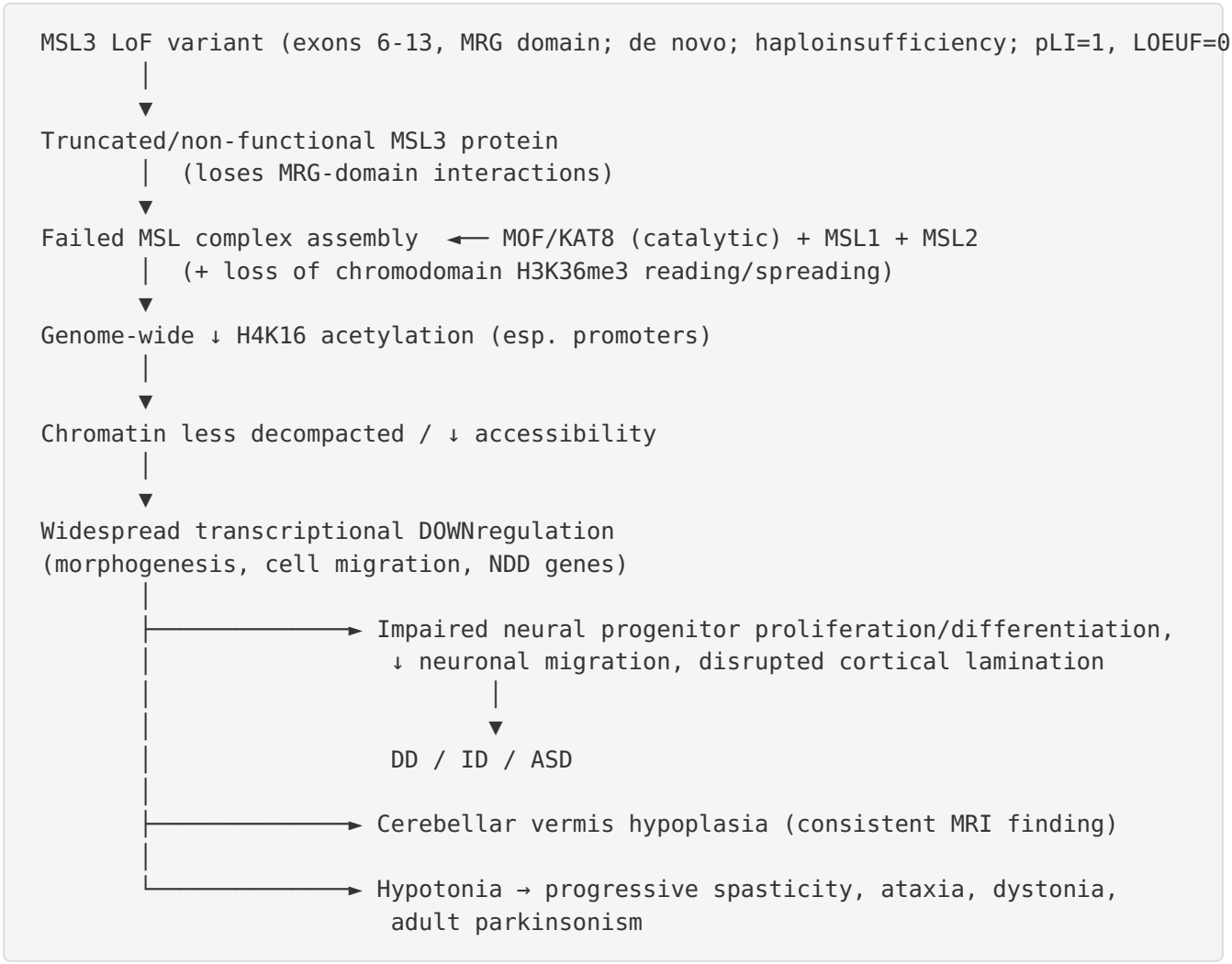
14. Other Species / Natural Disease

No naturally occurring companion-animal or wildlife disease is documented. The gene is deeply conserved: **orthologs** include *Drosophila melanogaster* msl-3 (dosage-compensation protein) and *S. cerevisiae* EAF3 (NuA4 subunit). Human MSL3: NCBI Gene 10943. No zoonotic or cross-species transmission (monogenic developmental disorder).

15. Model Organisms

- **Mouse:** nervous-system-specific **Msl2 conditional knockout** — the closest in vivo disease model; recapitulates reduced promoter H4K16ac, disrupted neocortical lamination, impaired neural progenitor proliferation/differentiation, reduced neuronal migration, and autism-like behaviors (PMID: 42168661). (Models the complex partner rather than MSL3 directly — a limitation.)
- **Human cellular models:** patient-derived lymphoblastoid/fibroblast lines and HEK293 transfection — demonstrate lost MOF/MSL1 interaction, reduced bulk H4K16ac, and transcriptome dysregulation (PMID: 30224647).
- **Drosophila:** msl-3 — chromodomain H3K36me3 reader; dosage compensation (PMID: 19029895).
- **Yeast:** EAF3/NuA4 homolog (PMID: 11036083).

Mechanistic Model / Interpretation



The model is **upstream = chromatin/H4K16ac defect**, **downstream = transcriptional and neurodevelopmental consequences**. The HDAC-inhibitor rescue in patient cells is the strongest evidence that the acetylation imbalance is causal and potentially reversible at the molecular level; the Msl2-cKO mouse provides the in vivo tissue-level causal link to neurodevelopmental phenotypes.

Evidence Base

PMID	Title (abbrev.)	Role in this report
30224647	*De novo mutations in MSL3 cause an X-linked syndrome…*	Syndrome-defining; H4K16ac loss; HDACi rescue; core clinical features
33173220	*Defining the genotypic and phenotypic spectrum of X-linked MSL3-related disorder*	25-patient cohort; variant spectrum/ clustering; frequencies; surveillance
40767387	*Prolonged follow-up in a 30-year-old male with MSL3 variant*	Case count (~42); adult survival; levodopa response
41216464	*Two Chinese patients with Basilicata-Akhtar syndrome*	MRG-domain LoF intolerance; rhGH treatment for GHD
42880250	*Molecular mechanism of residue-specific nucleosome acetylation*	MSL "molecular ruler"; H3K36me3 accelerates catalysis
42168661	*MSL2 disruption leads to autism-like behaviors…*	In vivo mouse model; reduced promoter H4K16ac; neurodevelopmental phenotypes
19029895	*The MSL3 chromodomain directs a key targeting step…*	Chromodomain reads H3K36me3; spreading function
17936709	*MSL complex attracted to H3K36me3-marked genes*	MSL3 binds H3K36me3 nucleosomes; sequence-independent targeting
11036083	*Yeast NuA4 and Drosophila MSL complexes…*	Evolutionary conservation (EAF3 homolog)

Additional supporting literature on MSL/MOF biology and H4K16ac mechanism: PMID: 38407474, PMID: 38790268, PMID: 20018852, PMID: 22421046, PMID: 19648925, PMID: 22821985.

Limitations and Knowledge Gaps

- **Small case numbers (~42 total):** frequency estimates, penetrance, expressivity, and natural history are based on a single ~25-patient cohort plus scattered case reports; prevalence/incidence are undefined.
- **No MSL3-specific animal model:** the strongest in vivo evidence comes from a **Msl2** conditional-knockout mouse, which models the complex partner rather than MSL3 itself; an MSL3-specific mouse recapitulation is not established.
- **Mechanistic inference for CNS anatomy:** the link from H4K16ac loss to cerebellar vermis hypoplasia specifically is inferred, not directly demonstrated in human tissue.
- **Alternative-transcript compensation hypothesis** (exon-clustering explanation) is plausible but not experimentally proven.
- **No therapeutic clinical trials:** HDAC-inhibitor benefit is demonstrated only in patient cells; rhGH and levodopa evidence is anecdotal (single patients/small series).
- **Orphanet/ICD-11 mappings** were not individually verified in this investigation.
- **No disease-specific mortality/QoL instrument** data exist.

Proposed Follow-up Experiments / Actions

1. **Generate an MSL3-specific conditional-knockout (and MRG-domain truncation knock-in) mouse** to directly test cerebellar vermis hypoplasia and cortical lamination phenotypes attributable to MSL3 (vs. Msl2).
2. **iPSC-derived cortical and cerebellar organoids** from patient lines to map H4K16ac loss, transcriptomic dysregulation, and migration/lamination defects in a human developmental context (single-cell multi-omics: scRNA-seq + CUT&Tag for H4K16ac + scATAC-seq).
3. **Preclinical HDAC-inhibitor efficacy and safety studies** in patient iPSC-neurons and in the mouse model, with careful attention to developmental timing/critical window and dose, to de-risk a potential clinical trial.
4. **Test the alternative-transcript compensation hypothesis** by quantifying the proposed N-terminal MSL3 isoform in relevant tissues and by modeling first-five-exon vs. terminal-exon truncations.

5. **Establish an international patient registry / natural-history study** to refine prevalence, penetrance, expressivity, movement-disorder trajectory (including adult parkinsonism), and standardized QoL outcomes.
6. **Functional classification pipeline for missense/VUS** (complex-assembly and H4K16ac assays) to support ACMG reclassification and improve diagnostic yield.
7. **Systematic endocrine screening (GH axis)** across the cohort to determine how often rhGH is indicated and to formalize growth surveillance guidelines.

Report compiled from an autonomous multi-iteration literature and database investigation. Evidence is human clinical (patient cohorts/case reports), model organism (mouse Msl2-cKO, Drosophila, yeast), and in vitro (patient-derived cells), as indicated.
